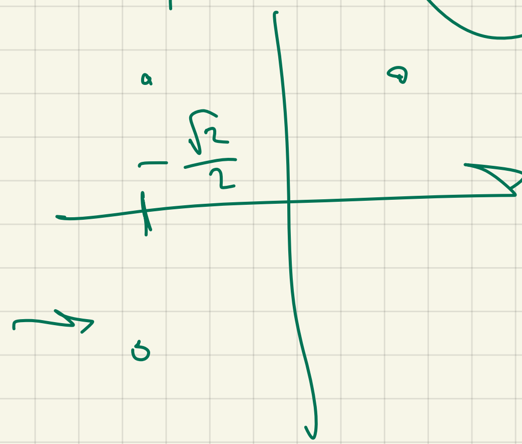
$$\sqrt{72}-\sqrt{288}\sin^2\frac{21\pi}{8}$$

$$\sqrt{72}\left(1-2\sin^2\frac{21\pi}{8}\right)=\sqrt{72}\cdot\left(-\frac{\sqrt{2}}{2}\right)=-6$$

$$\sqrt{72}\cos\frac{21\pi}{4}$$

$$\frac{5\pi}{4}-2\pi=\frac{5\pi-8\pi}{4}=-\frac{3\pi}{4}$$

$$\frac{5\pi}{4}$$



$$k+\frac{a-kb}{x+b}$$

$k=-2$

$b=-4$

$(1;-3)$

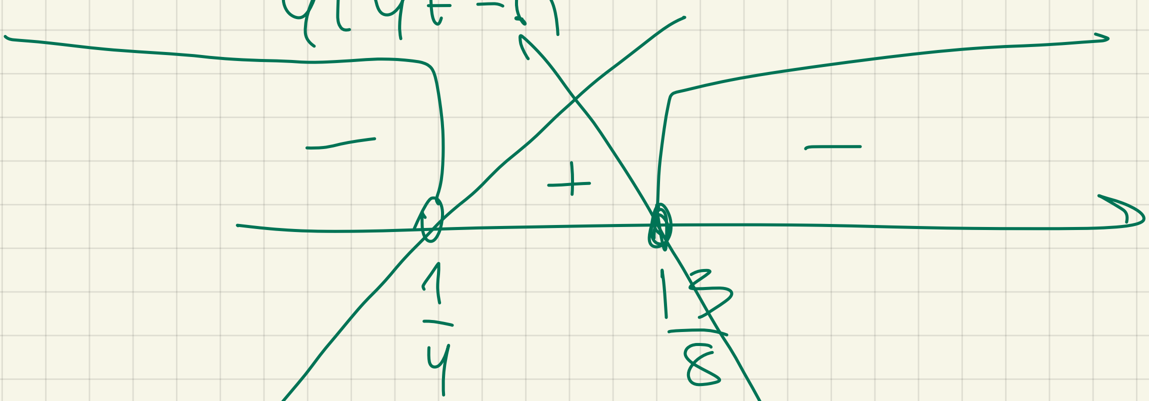
$$\frac{3-(x+5)^{-1}}{4(x+5)^{-1}-1} \le \frac{1}{4}$$

$$(x+5)^{-1} = t$$

$$\frac{3-t}{4t-1} \le \frac{1}{4}$$

$$\frac{12-4t-4t+1}{4(4t-1)} \le 0$$

$$\frac{-8t+13}{4(4t-1)} \le 0$$

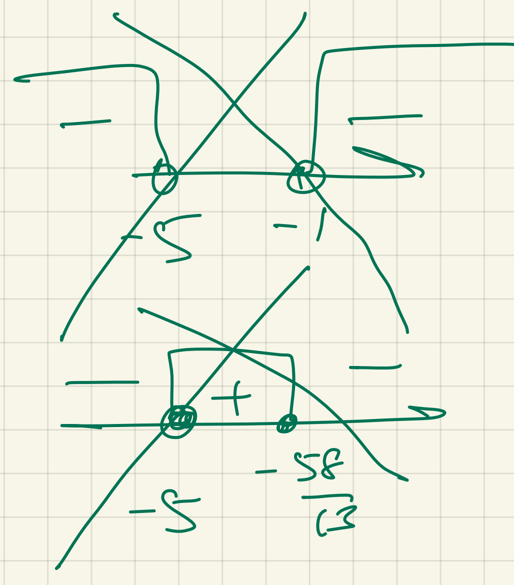


$$\left[\begin{array}{l} t < \frac{1}{4} \\ t \ge \frac{13}{8} \end{array}\right. \Rightarrow \left[\begin{array}{l} (x+5)^{-1} < \frac{1}{4} \\ (x+5)^{-1} \ge \frac{13}{8} \end{array}\right.$$

$$\left[\begin{array}{l}\dfrac{1}{x+5} < \dfrac{1}{4} \\[2ex] \dfrac{1}{x+5} \geq \dfrac{13}{8}\end{array}\right. \Rightarrow$$

$$\left[\begin{array}{l}\dfrac{4-x-5}{4(x+5)} < 0 \\[2ex] \dfrac{8-13x-65}{8(x+5)} \geq 0\end{array}\right.$$

$$\left[\begin{array}{l}\dfrac{-1-x}{4(x+5)} < 0 \\[2ex] \dfrac{-58-13x}{8(x+5)} \geq 0\end{array}\right.$$



$S = 4\,420\,000$р $\quad 10\%$

		пл.	ост
1.	$S \cdot 1{,}1$	0	$1{,}1S$
2.	$1{,}1^2 S$	x	$1{,}1^2 S - x$
3.	$1{,}1^3 S - 1{,}1x$	0	$1{,}1^3 S - 1{,}1x$
4.	$1{,}1^4 S - 1{,}1^2 x$	x	$1{,}1^4 S - 1{,}1^2 x - x$

$$1{,}1^4 S - 1{,}1^2 x - x = 0$$

16

+16

$$x = \frac{1{,}1^4 S}{1 + 1{,}1^2} = \frac{1{,}1^4}{2{,}21} \cdot 4\,420\,000$$

2 928 200р

$$x = a(x+1)$$

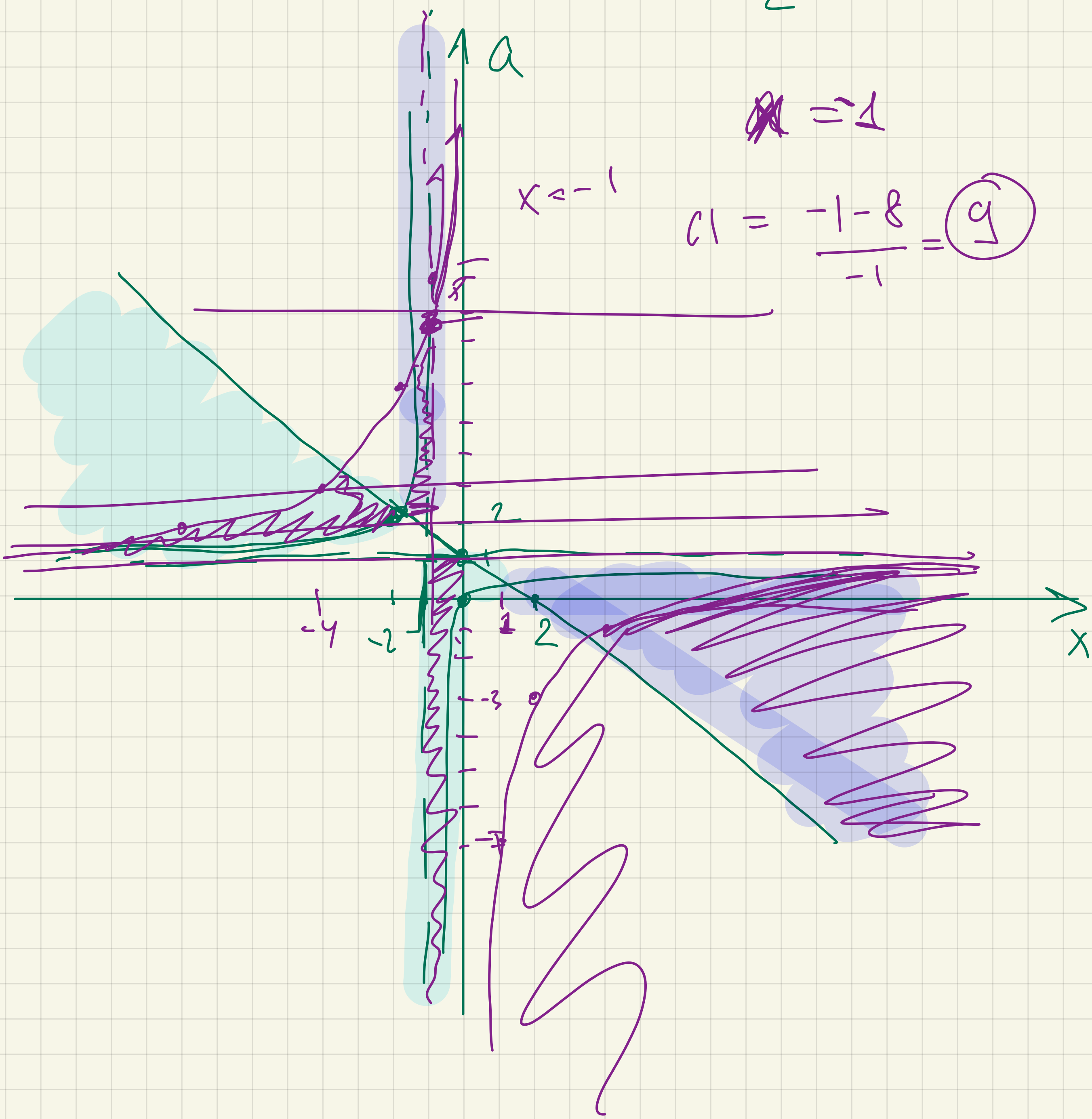
$$\begin{cases} \dfrac{x - ax - a}{x + 2a - 2} \geq 0 \\ x - ax > 8 \end{cases}$$

$$a = \frac{x}{x+1} =$$

$$\frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$$

$$a < \frac{x-8}{x} = 1 - \frac{8}{x}$$

$$\left[\begin{array}{l}\begin{cases} x - ax - a \geq 0 \\ x + 2a - 2 > 0 \end{cases} \\ \begin{cases} x - ax - a \leq 0 \\ x + 2a - 2 < 0 \end{cases}\end{array}\right. \to \begin{array}{l}\begin{cases} a \leq 1 - \frac{1}{x+1} \\ a > \frac{2-x}{2} \end{cases} \\ \begin{cases} a \geq 1 - \frac{1}{x+1} \\ a < \frac{2-x}{2} \end{cases}\end{array}$$



$\cancel{a} = 1$

$x < -1$

$a = \frac{-1-8}{-1} = 9$

$$\begin{cases} \log_{2023}\left(1012+1011\dfrac{|x|}{x}\right)+(x-y)^2=a \\ \log_{2024}(3-y)=0 \end{cases}$$

2 реш

$3-y>0 \quad y<3$

$3-y=2024^0=1$

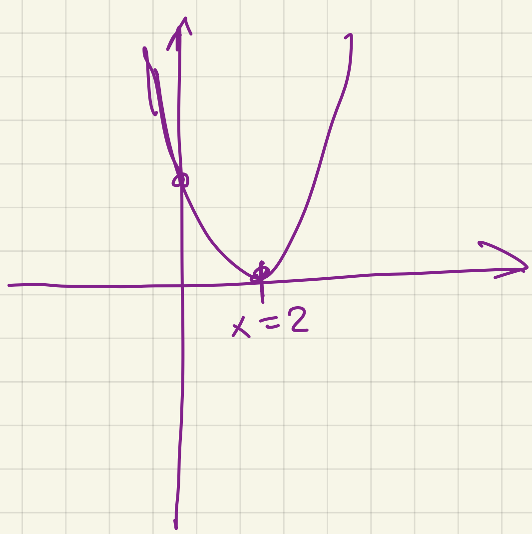
$y=2$

$$\log_{2023}\left(1012+1011\frac{|x|}{x}\right)+(x-2)^2=a$$

2 реш

$x \geq 0 \qquad x<0$

$x<0 \Rightarrow \qquad 0+(x-2)^2=a$



$$\begin{cases} x^4 - y^4 = 12a - 28 \\ x^2 + y^2 = a \end{cases}$$

$\boxed{\text{4 реш}}$

$$\begin{cases} (x^2 - y^2)(x^2 + y^2) = 12a - 28 \\ x^2 + y^2 = a \end{cases}$$

$$\begin{cases} (x^2 - y^2)\,a = 12a - 28 \\ x^2 + y^2 = a \end{cases}$$

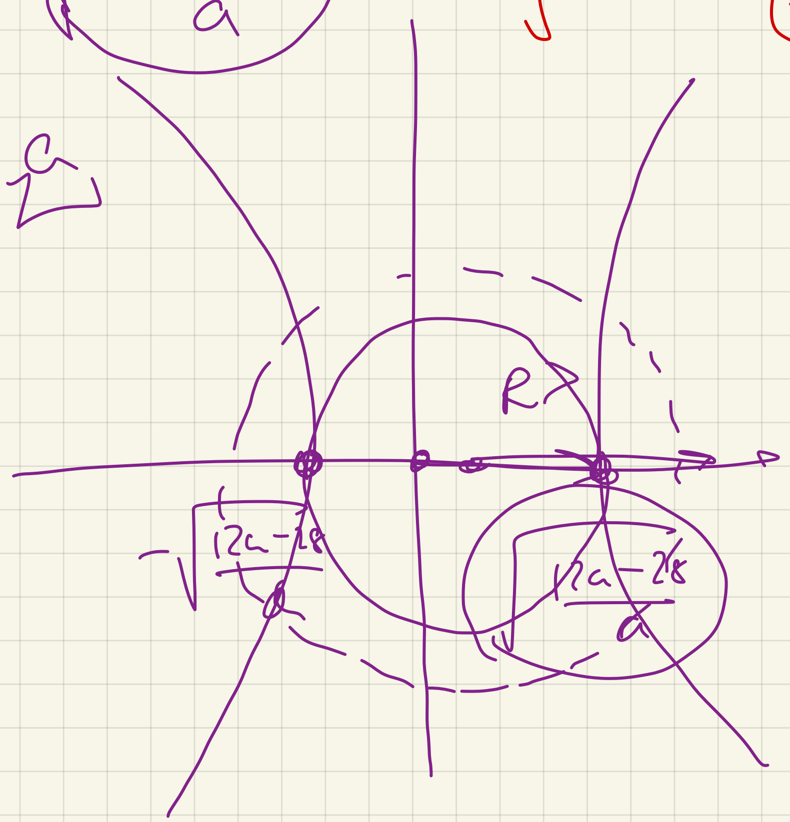
$$\begin{cases} x^2 - y^2 = \dfrac{12a - 28}{a} \\ x^2 + y^2 = a \end{cases}$$

$+$

$-$

$$2x^2 = \frac{12a-28}{a} + a$$

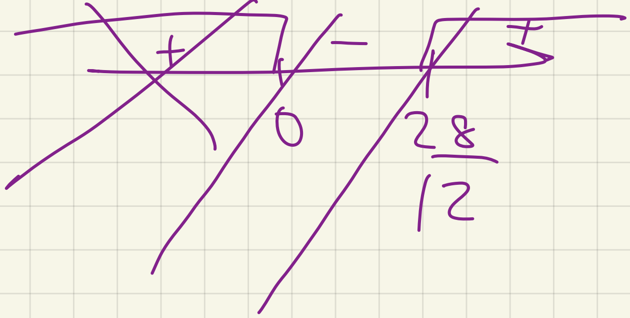
$$2y^2 = a - \left(\frac{12a-28}{a}\right)$$



$R = \sqrt{a}$

$l = \boxed{\sqrt{\dfrac{12a-28}{a}}}$

$$\frac{12a-28}{a}$$



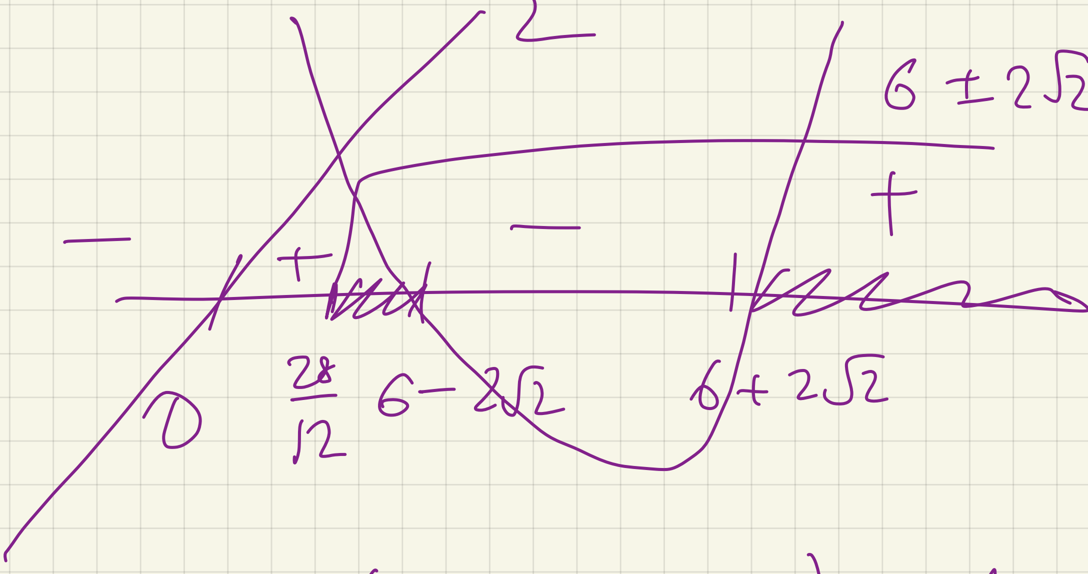
$$R > \ell$$

$$a > \frac{12a-28}{a}$$

$$\frac{a^2-12a+28}{a} > 0$$

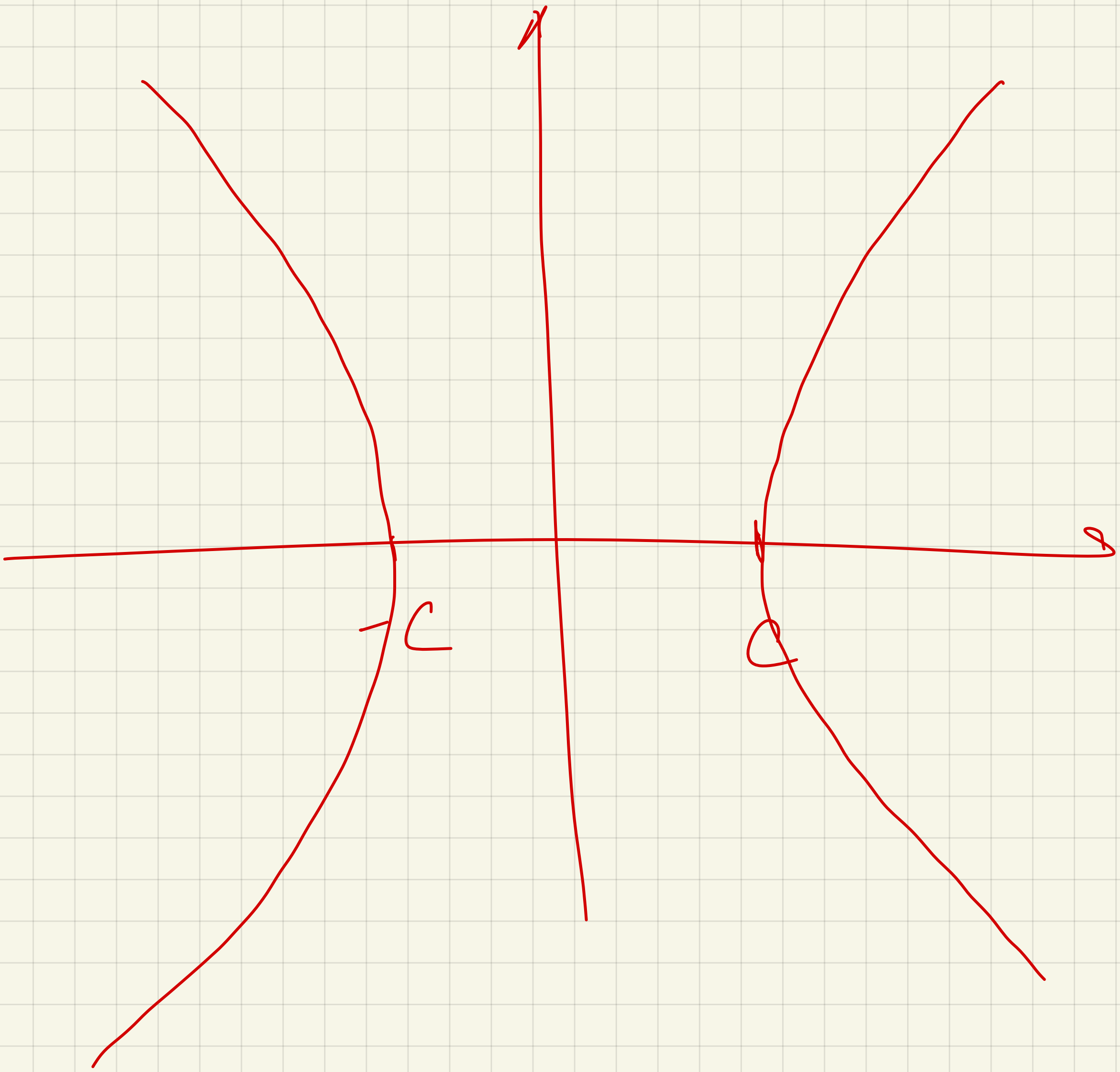
$$\frac{12 \pm \sqrt{12^2 - 28 \cdot 4}}{2} = \frac{12 \pm 4\sqrt{2}}{2} =$$

$$6 \pm 2\sqrt{2}$$



$$a \in \left(\frac{28}{12} ; 6-2\sqrt{2}\right) \cup \left(6+2\sqrt{2} ; +\infty\right)$$

$$x^2 - y^2 = c^2$$



$$\begin{cases} x^2 - y^2 - 2(x+2y) - 3 = 0 \\ x^2 + y^2 - 2a(x+y) - 2x - 2a^2 + 6a \leq 0 \end{cases}$$

1 реш

$(x-1-a)^2 + (y-a)^2 \le 4a^2 - 4a + 1$

$(1+a,\ a)$

$(2a)^2 - 2 \cdot 2a + 1$

$(2a-1)^2$

$y = x - 3$

$y = -x - 1$

$R = |2a - 1|$